The Sign Test

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Introduction

The sign test is the most assumption-light procedure for a one-sample median or paired comparison. It ignores the magnitude of deviations and uses only whether each observation is above or below the reference value (or whether each pair increased or decreased).

Prerequisites

Binomial distribution, paired data.

Theory

For iid $X_1, \ldots, X_n$ and a hypothesised median m_0, let $S = \#\{X_i > m_0\}$. Under H_0: median $= m_0$,

$$S \sim \text{Binomial}(n', 0.5),$$

where n' excludes any $X_i = m_0$. The two-sided p-value is computed from the binomial PMF.

For paired data, apply the test to the signs of the paired differences.

The sign test has lower power than the Wilcoxon signed-rank when the symmetry assumption of the latter holds. Its advantage is that it works under **any** continuous distribution.

Assumptions

- iid observations (or independent pairs).
- Continuous distribution (else need to handle ties at the reference).

No symmetry, no shape assumption.

R Implementation

```r
library(BSDA)
set.seed(2026)

# One-sample: test whether median equals 100
x <- rlnorm(30, meanlog = log(105), sdlog = 0.3)
SIGN.test(x, md = 100)

# Paired: test whether intervention changes a score
pre  <- rpois(25, 5)
post <- pre + sample(-1:3, 25, replace = TRUE)
SIGN.test(post, pre, md = 0)
```

Output & Results

```
        One-sample Sign-Test
data:  x
s = 21, p-value = 0.021
alternative hypothesis: true median is not equal to 100
95 percent confidence interval: 97.5 to 120

        Dependent-samples Sign-Test
data:  post and pre
S = 19, p-value = 0.016
95 percent CI for median difference: 0.5 to 2.0
```

Both tests reject: the one-sample median significantly exceeds 100; paired differences are significantly positive.

Interpretation

"The median post-intervention score was 7.0 (95 % CI 0.5 to 2.0 for the paired difference), significantly higher than pre (sign test, 19 of 25 positive differences, exact binomial $p = 0.016$)."

Practical Tips

- Use the sign test when the distribution of paired differences is badly skewed or clearly not symmetric.
- Report the number of positive, negative, and zero differences; the latter are excluded from the test.
- The sign test's 95 % CI for the median is a distribution-free interval based on order statistics.
- With many ties at the reference, consider the Wilcoxon or rank-based alternative, understanding their stronger assumptions.
- Power is lower than Wilcoxon signed-rank; prefer Wilcoxon when its symmetry assumption is plausible.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Populations, samples, and the central limit theorem
- Bootstrap and permutation tests
- Maximum likelihood estimation
- One-sample t-test and one-proportion test
- Hypothesis testing, p-values, type I/II errors
- Two-sample and paired t-tests
- Two proportions, chi-square, risk and odds
- Pearson and Spearman correlation
- Non-parametric tests